



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



Microsoft Azure AI Fundamentals (AI-900)

TGS Ref No: TGS-2023021100

Version 1.0

Lesson Plan - Microsoft Azure AI Fundamentals (AI-900)

TGS-2023021100 - Version 1.0 - 8 July 2026

Course Information

Field	Detail
Duration	2 Days / 16 training hours, 9:00 AM - 6:00 PM
Delivery Mode	Instructor-led with lab-based guided practice
Assessment	Open-book written / MCQ assessment on LMS/TMS
Trainer Preparation	Verify slides, Learner Guide, lab repository, LMS/TMS, Azure access, practice exam link, and TRAQOM QR codes.
Course Repository	github.com/tertiarycourses/TGS-2023021100---Microsoft-Azure-AI-Fundamentals-AI-900-

Delivery Approach

- Use short concept briefings followed by guided lab practice.
- For each lab, ask learners to identify input, output, Azure service, validation evidence, and responsible AI control.
- Use checkpoints for remediation before moving to the next topic.
- Keep attendance, assessment, TRAQOM, cleanup, and practice exam actions visible near the end of Day 2.

Detailed Lesson Schedule

Day 1 - AI and Machine Learning Foundations

Time	Lesson Focus	Trainer Notes	Learner Evidence	Slides
9:00-9:15	Course opening and attendance	Confirm TRAQOM attendance, class logistics, assessment rules, courseware access, and learner expectations.	Learners scan attendance, confirm access to slides/LG/labs, and state role or AI experience.	1-23
9:15-9:45	AI workloads and responsible AI	Explain workload categories with workplace examples; prompt learners to classify each scenario by input, output, and risk.	Learners classify scenarios and record responsible AI questions for later lab checkpoints.	24-30
9:45-10:30	Lab 01: AI workloads and responsible AI	Guide learners through workload recognition and responsible AI review; pause after each checkpoint for discussion.	Completed Lab 01 notes: workload selected, responsible AI issue identified, and validation response prepared.	31-34
10:30-10:45	Tea break	Remind learners to save lab notes and keep Azure/Microsoft Learn sessions active.	Break.	-
10:45-11:30	Lab 02: Machine learning fundamentals	Demonstrate features, labels, training data, evaluation, and prediction vocabulary before hands-on practice.	Learners explain the difference between classification, regression, clustering, and model evaluation.	35-38
11:30-12:15	Lab 03: Azure Machine Learning	Walk through Azure Machine Learning/AutoML concepts and connect portal steps to the ML lifecycle.	Learners identify model lifecycle stages and note where evaluation and deployment evidence appears.	39-42
12:15-1:00	Guided review: ML lifecycle and service mapping	Use the services map to compare Azure Machine Learning with prebuilt Azure AI services.	Learners complete a mini service-selection exercise and justify the service chosen.	24-30
1:00-2:00	Lunch break	Confirm afternoon lab sequence and support learners who need account access help.	Break.	-
2:00-2:45	Lab 04: Computer vision, image analysis, OCR	Brief image inputs, object/text extraction, confidence scores, and human review before the lab.	Learners validate image/OCR output and list accuracy or privacy risks.	43-46
2:45-3:30	Lab 05: Natural language processing	Explain sentiment, key phrase extraction, entity recognition, and language service boundaries.	Learners compare NLP output with the source text and capture a review question.	47-50
3:30-3:45	Tea break	Check progress and identify learners needing remediation before the workshop.	Break.	-
3:45-4:30	Scenario workshop	Facilitate small-group service selection across image, text, and ML scenarios.	Learners produce a workload-service-control mapping for at least three scenarios.	24-30, 43-46-47-50
4:30-5:15	Checkpoint and remediation	Review wrong answers, address misunderstood vocabulary, and connect answers back to lab evidence.	Learners update lab notes and complete checkpoint questions.	Lab review slides
5:15-6:00	Day 1 recap and Q&A	Summarize workload recognition, ML lifecycle, vision, and language services; assign revision topics.	Learners state one concept mastered and one concept to review on Day 2.	Review slides

Day 2 - Azure AI Services, Generative AI, and Assessment

Time	Lesson Focus	Trainer Notes	Learner Evidence	Slides
9:00-9:15	Attendance and Day 1 recap	Confirm AM attendance and run a quick recap quiz on workload-service mapping.	Learners answer recap questions and reopen their lab repository/courseware.	1-23

Time	Lesson Focus	Trainer Notes	Learner Evidence	Slides
9:15-10:00	Lab 06: Speech, translation, conversational AI	Explain speech-to-text, text-to-speech, translation, and conversation use cases before lab work.	Learners validate transcript/translation outputs and record accessibility or privacy considerations.	51-54
10:00-10:45	Lab 07: Document intelligence and search	Demonstrate document extraction, knowledge mining, indexing, and search concepts.	Learners identify extracted fields, indexing value, and when human review is required.	55-58
10:45-11:00	Tea break	Check lab progress and prepare generative AI safety framing.	Break.	-
11:00-12:00	Lab 08: Generative AI, Foundry, Azure OpenAI	Introduce prompts, grounding, content safety, model catalog, and review controls.	Learners test prompts, evaluate responses, and document grounding/review safeguards.	59-62
12:00-1:00	Guided practice: generative AI controls	Facilitate comparison of prompt-only, grounded, and reviewed outputs; highlight risk controls.	Learners refine prompts and write a short governance note for generative AI use.	24-30
1:00-2:00	Lunch break	Confirm remaining lab and assessment timing.	Break.	-
2:00-2:45	Lab 09: solution design and governance	Coach learners through service selection, RBAC/privacy/cost controls, and cleanup decisions.	Learners complete a solution design map and name at least three controls.	63-66
2:45-3:30	Lab 10: capstone and review	Guide the capstone service map and 7-day revision plan; validate all lab evidence is complete.	Learners submit/review the capstone map and identify personal revision priorities.	67-70
3:30-3:45	Assessment briefing	Review allowed materials, digital attendance, LMS/TMS submission, timing, and conduct rules.	Learners prepare approved materials and complete assessment attendance when instructed.	19, 20, 73-74
3:45-5:00	Final assessment	Invigilate the written/MCQ assessment, support LMS/TMS technical access only, and enforce assessment rules.	Learners complete assessment and submit answers on LMS/TMS.	19, 20, 73-74
5:00-5:15	Tea break	Collect any operational issues and prepare closing checks.	Break.	-
5:15-6:00	Course close, TRAQOM, practice exam	Confirm TRAQOM, final attendance, certification actions, resource cleanup, and practice exam access.	Learners complete required surveys/attendance and save the practice exam link for revision.	73-75

Lab Alignment and Evidence

Use this matrix to confirm every lab produces evidence linked to the lesson outcome and slide numbers.

Lab	Lab Title	Slides	Trainer Focus	Learner Evidence / Checkpoint
01	AI Workloads and Responsible AI	31-34	AI-900-style questions often ask you to identify workload types and responsible AI principles from a scenario.	You should have workload examples, Azure service mapping, and a responsible AI checklist. Checkpoint: What is a computer vision workload?; Why is fairness important?
02	Machine Learning Fundamentals	35-38	Know how to select the basic ML technique from the wording of a business scenario.	You should have scenario mappings, feature/label examples, and a training/validation explanation. Checkpoint: What is regression used for?; What is classification used for?
03	Azure Machine Learning, Automated ML, Model Lifecycle	39-42	Understand what Azure Machine Learning does at a high level: data, compute, training, automated ML, model management, deployment, and monitoring.	You should have a workspace area map, AutoML explanation, and model lifecycle flow. Checkpoint: What is automated machine learning?; What is a model endpoint?
04	Computer Vision, Image Analysis, OCR	43-46	Vision questions often ask which workload or Azure service fits the scenario.	You should have a scenario mapping table, Azure AI Vision capability notes, and responsible AI review. Checkpoint: What is OCR?; How is object detection different from image classification?
05	Natural Language Processing and Azure AI Language	47-50	NLP questions often describe text, speech, translation, sentiment, or entity extraction scenarios.	You should have NLP workload mapping, Azure AI Language notes, and a feedback pipeline design. Checkpoint: What is sentiment analysis?; What is entity recognition?
06	Speech, Translation, Conversational AI	51-54	Speech and translation questions usually test whether you can identify the workload and the right Azure AI service.	You should have scenario mappings, Azure service notes, support assistant flow, and responsible AI review. Checkpoint: What is speech recognition?; What is speech synthesis?
07	Document Intelligence and Knowledge Mining	55-58	Document processing questions may reference OCR, forms, invoices, extraction, and search.	You should have a document task map, extraction flow, knowledge mining notes, and responsible AI review. Checkpoint: What is document processing?; How is OCR related to document intelligence?
08	Generative AI, Azure AI Foundry, Azure OpenAI, Model Catalog	59-62	Current AI fundamentals material emphasizes generative AI, Azure AI Foundry, Azure OpenAI, and model catalog concepts.	You should have a generative AI glossary, scenario map, Azure capability notes, and safe assistant design. Checkpoint: What is generative AI?; What is a prompt?
09	AI Solution Design, Security, Cost, Governance	63-66	AI solution questions often test choosing the right service and recognizing responsible AI, privacy, and security concerns.	You should have a requirements summary, service selection table, security controls, cost controls, and responsible AI controls. Checkpoint: Why does AI solution design include governance?; What is RBAC used for?
10	AI-900 Capstone and Course Review	67-70	The goal is to recognize AI workloads, understand foundational concepts, and describe how Azure AI services support common business scenarios.	You should have a service map, responsible AI action list, ML technique table, review log, and cleanup confirmation. Checkpoint: Which AI workload is easiest for you to identify?; Which Azure service names are most confusing?

Assessment and Funding Requirements

- Minimum 75% attendance across the course based on SSG digital attendance records.
- Complete assessment digital attendance before final assessment.
- Use approved course materials only for the open-book written / MCQ assessment.
- Submit answers through LMS/TMS and sign the Assessment Summary Record when instructed.
- Complete feedback and TRAQOM survey; use exams.tertiaryinfotech.com for revision.

Trainer Resource Checklist

Item	Trainer Check
Slides and Learner Guide	Confirm latest courseware files are in the courseware folder and on LMS/TMS.
Lab repository	Confirm learners can access github.com/tertiarycourses/TGS-2023021100---Microsoft-Azure-AI-Fundamentals-AI-900- .
Azure access	Confirm subscription/sandbox availability and portal sign-in.
Assessment	Prepare LMS/TMS assessment link, attendance QR, and Assessment Summary Record process.
Cleanup	Remind learners to delete or stop personal Azure resources.